

Soheil Sepahyar, Ph.D.

Department of Computer Science, University of Massachusetts Boston

soheil.sepahyar@umb.edu — (906) 370-0091

[linkedin.com/in/soheil-sepahyar](https://www.linkedin.com/in/soheil-sepahyar)

Academic Appointment

Lecturer, Department of Computer Science, University of Massachusetts Boston 2024–Present

Education

Ph.D., Computer Science, Michigan Technological University 2019–2023

Dissertation: *Distance Perception in Virtual Reality* ([link](#))

Advisor: Prof. Scott Kuhl

M.S., Computer Science, Michigan Technological University 2019–2022

B.S., Computer Software Engineering, Azad University, Tehran 2014–2018

Research Interests

Virtual and Augmented Reality (VR/AR); Human–Computer Interaction; neural rendering and foveated perception;

Artificial neural networks and deep learning; robot learning (imitation learning, diffusion policies, sim-to-real);

Computer graphics and digital-twin robotics simulation in NVIDIA Isaac Sim.

Current Projects

- **AI-driven digital-twin robotics in Isaac Sim with VR/MR data collection.** Building a VR/MR teleoperation pipeline in NVIDIA Isaac Sim to collect multi-modal trajectories (head and hand poses, RGB-D, gaze, actions) and to train neural control policies. Methods include behavior cloning, diffusion policies, and offline reinforcement learning for manipulation and navigation, plus representation learning for sim-to-real transfer.
- **Dynamic foveated rendering for distance judgment in VR.** Real-time, depth-aware, gaze-based rendering using Meta Quest Pro eye tracking in Unity with custom shaders. Uses vergence to keep foveal regions sharp while applying peripheral blur. Planned AI extension: a gaze-conditioned upsampler and peripheral reconstruction model to improve image quality, comfort, and task performance.

Academic and Research Experience

Lecturer, University of Massachusetts Boston 2024–Present

- Teach and coordinate large-enrollment undergraduate courses (approximately 150–200 students per semester) across multiple sections, including:
 - CS110 - Introduction to Computing
 - CS210 - Intermediate Computing with Data Structures

- CS220 - Applied Discrete Mathematics
- CS240 - Programming in C
- CS410 - Software Engineering
- CS420 - Theory of Computation
- CS450 - Structure of Higher-Level Programming Languages
- Design and deliver structured curricula, programming assignments, and exams with emphasis on conceptual clarity, problem-solving, and real-world applicability across both introductory and advanced courses.
- Implement scalable assessment systems using Gradescope and custom autograding pipelines, improving grading efficiency and consistency across large cohorts.
- Foster an inclusive and supportive learning environment by maintaining clear communication, structured course organization, and responsive student support.
- Manage course logistics and coordination across multiple sections, ensuring consistency in instruction, grading, and student experience.

Ph.D. Researcher, Michigan Technological University 2019–2023

- Conducted experimental research on distance perception in virtual environments using Unity and OpenGL.
- Published and presented findings at the ACM Symposium on Applied Perception (SAP 2020, 2022).

Industry Experience

Product Design Technical Lead Intern, Visteon Corporation, Michigan Summer 2022

- Developed a Python-based analytics suite (V-PEDAT) to process validation data and generate automated reports.
- Reduced manual analysis time by over 90% and improved team efficiency across global validation groups.

AI and System Integration Intern, Visteon Corporation, Michigan Summer 2021

- Optimized driver-interaction detection pipelines using sensor fusion and geometric analysis.
- Benchmarked embedded inference performance and supported ECU integration.

ADAS Research Intern, Visteon Corporation, Michigan Summer 2020

- Automated LiDAR dataset processing using CUDA and Docker pipelines.
- Assisted with dataset preparation and validation for ADAS perception models.

Publications

- [1] Sepahyar, S. et al. “*VR Distance Judgements.*” *ACM Symposium on Applied Perception (SAP)*, 2022. [link](#)
- [2] Sepahyar, S. et al. “*Brightness vs. Distance Judgements.*” *ACM SAP*, 2020. [link](#)
- [3] Sepahyar, S. et al. “*Sorting Algorithm Comparison.*” *ACM ACAI*, 2019. [link](#)

In Preparation

- Sepahyar, S., Kuhl, S. Impact of Pre-experiment Walking on Distance Judgments in VR. Target journal: *ACM Transactions on Applied Perception (TAP)*.

Awards and Honors

Finishing Fellowship Grant - Michigan Technological University (2023)

Business Opportunity Recognition Certificate - VRSPACE (2023)

Professional Service

Reviewer: IEEE VR (2023–2025), VRST (2023–2025)

Program Committee Member: ACM Symposium on Applied Perception (2025, 2026)

Teaching Interests

Programming fundamentals, data structures, theory of computation, discrete mathematics, computer graphics, human–computer interaction, virtual reality, artificial intelligence, deep learning.

Technical Skills

C, C++, Python, C#, Java, Racket, Rocq (Coq) Unity, OpenGL, GLSL, CUDA;
PyTorch, TensorFlow; NVIDIA Isaac Sim and Omniverse; Git, Docker, CMake; Linux, Windows.

Citizenship

U.S. Permanent Resident (Green Card)